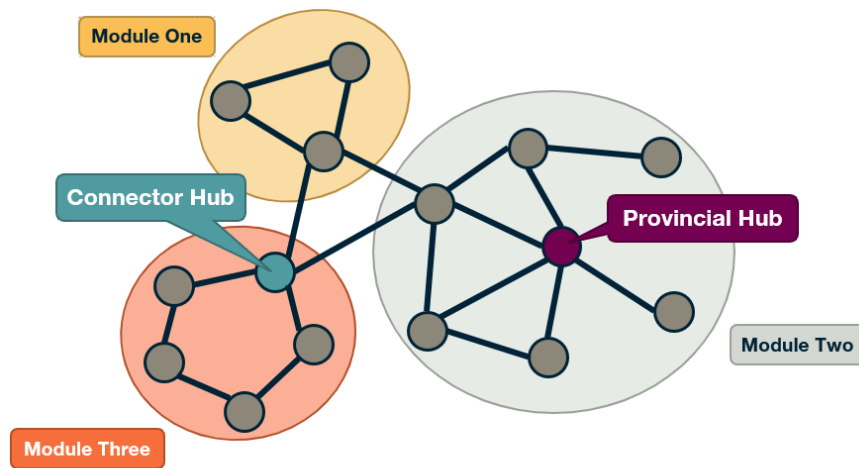


L8: Participation Coefficient



An interesting question is to examine the role of individual nodes within the community they belong to.

Is a node mostly connected to other nodes of the same community? Or is it connected to several other communities, acting as a “**bridge**” between its own community and others? A commonly used metric to answer such questions is the **Participation Coefficient** of a node, measuring

the participation of that node in *other* communities.

Suppose that we have already discovered that the network has n_c communities (or *modules*). Also, let $k_{i,s}$ be the number of links of node i that are connected to nodes of community s , and k_i the degree of node i .

The participation coefficient is then defined as:

$$P_i = 1 - \sum_{s=1}^{n_c} \left(\frac{k_{i,s}}{k_i} \right)^2$$

Note that if all the connections of node i are with other nodes in its own community, we have that $P_i = 0$. An example of such a node is the highlighted node at module-two in the visualization.

On the other hand, if the connections of node i are uniformly distributed with all n_c communities, the participation coefficient will be close to 1. This is the case with the highlighted node at module-three in the visualization.

The (loose) upper bound $P_i = 1$ is approached for nodes with very large degree k_i , if their connections are uniformly spread to all n_c communities.

Food for Thought

What is the rationale for the “squaring operation” at the previous formula? Why not just take the summation of those fractions?